Fiscal Year:	FY 2004	**Task Last Updated:**	FY 05/21/2004
PI Name:	Nagourney, Warren Ph.D.		
Project Title:	Ultrahigh Resolution Optical Frequency Standard Using Individual Indium Ions		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	warren@phys.washington.edu	**Fax:**	FY 206-685-0635
PI Organization Type:	UNIVERSITY	**Phone:**	206-543-9585
Organization Name:	University of Washington		
PI Address 1:	Physics Department		
PI Address 2:	Mail Stop 351560		
PI Web Page:			
City:	Seattle	**State:**	WA
Zip Code:	98195	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	04/20/2001	**End Date:**	02/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JPL
Contact Monitor:	Maleki, Lute	**Contact Phone:**	818-354-3688
Contact Email:	lute.maleki@jpl.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dehmelt, Hans (University of Washington) Fortson, Edward (University of Washington)		
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth"s gravity		

Task Description:	We propose to construct an optical frequency standard based upon a single, laser-cooled, trapped indium ion. The proposed standard, whose absolute accuracy should approach one part in 10^18, would benefit from the relatively vibration-free environment of space and from the increasing compactness and efficiency of narrowband sources of radiation in the UV region of the spectrum. Recently, terrestrial sources have demonstrated sub-Hertz linewidths over reasonable times, but the microgravity environment should make this easier to achieve. Recent advances in coherently relating an optical frequency to a microwave frequency should allow one to convert the frequency standard into a time standard without loss of accuracy. This eliminates the chief objection to the use of an optical frequency as a primary time standard: the difficulty in coherently relating a frequency of about 1e15 Hz to a microwave frequency which can be electronically counted and which therefore can be used to define the second. The advantages of optical standards with their much higher intrinsic Q and potentially smaller systematic errors (when compared to microwave standards) gives an optical standard a new lease on life as a promising candidate for the future primary time standard of choice. The task for the first year of the project will be to improve the linewidth of the recently-observed resonances of the forbidden transition to several 10's of Hz and the second year's task will be the construction of a compact, solid-state frequency-quadrupled cooling laser, a "twin" of the existing compact "clock" laser currently being used for spectroscopy. With two solid-state laser systems, the entire optical frequency standard could be quite compact and would consume a fairly small amount of electrical power; these qualities would be very favorable for the potential use of the standard in space. An extremely stable optical frequency standard has many possible applications in the areas of metrology, tests of general relativity and the equivalence principle, detection of gravitational waves as well as the more mundane areas of highly precise navigation and surveying. A frequency standard based upon a single indium ion could be used in a proposed experiment to measure the time rate of change of the fine structure constant. This has become particularly interesting due to the recent astronomical observation of a statistically significant change in alpha over a large fraction of the size of the universe. The proposed experiment would compare a single ion indium standard (the reference) to a single ion thallium standard, whose change of frequency with alpha is about 5 times greater than in indium. The sensitivity in the fractional change of alpha is anticipated to be 10^-16/year.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Improved time metrology will have many benefits for navigation, surveying and communication.
Task Progress:	We made a recent breakthrough in our highly stable "clock" laser by replacing the commercial high-power diode laser (MOPA) with a 946 nm YAG-based non-planar ring oscillator (NPRO). We formerly had high hopes for the MOPA, since it was relatively stable and generated high power, facilitating efficient generation of 236 nm "clock" radiation in two stages of enhanced frequency doubling. The MOPA was unfortunately a recently "orphaned" commercial product and had very restricted access to the diode current which would be used for frequency stabilization. It was therefore somewhat difficult to stabilize it to the ultra-stable optical cavity and a linewidth of only several kHz was obtained with difficulty. We were reluctant to "dig into" the laser in order to increase the bandwidth of the current driver since the laser is no longer supported by the manufacturer and a mistake could destroy the diode (and make the laser worthless). We were very pleased to recently see a commercial realization of the very stable YAG non-planar ring oscillator at 946 nm. It turns out that this approach was one we recommended 17 years ago when we first sought funding (unsuccessfully) for the indium experiment. Even though YAG oscillates at a limited number of discrete wavelengths, by a fortuitous circumstance, one of its lines (946 nm) is the exact fourth sub-harmonic of the "clock" transition in indium. The YAG NPRO is extremely stable and very simple to manufacture. A commercial (American) source for these lasers refused to make one at 946 nm and we therefore chose the MOPA for the indium "clock" laser. This state of affairs changed with the release of a high-power YAG NPRO by a German company early in 2004. This laser has sufficient power (about 300 mW) to generate a useful amount of power at 236 nm and it has an extraordinary 2 kHz free-running linewidth: more than 1000 times less than that of the MOPA! Moreover, it has convenient and fast (time constant 1 sec) temperature tuning for large frequency changes and a fast (resonant frequency 100 kHz) PZT for small frequency changes (and laser stabilization). The laser was purchased and introduced into our system early in 2004. Using a very simple feedback system and our highly-stable Zerodur optical cavity (finesse 30,000, drift 5 Hz/sec), we were able to immediately narrow the laser to under 100 Hz in the short term with a long-term drift of 5 Hz per second. This linewidth should be reduced considerably in the near future by modest improvements to the servo electronics. The frequency multiplication currently produces about 100 microwatts at 236 nm; this is more than adequate to drive the "clock" transition and significantly more is expected as we optimize the system. A sample of the blue light from the first doubling stage is delivered to a saturated tellurium spectrometer which is used to set the frequency of the laser with a reproducibility of about 100 kHz. The tellurium lines are quite strong near the subharmonic of the "clock" frequency and this frequency setting therefore works quite well (by comparison, a digital wavemeter would be from 10 to 100 times less precise). The Zerodur cavity is fixed in frequency and not tunable: we were able to tune the locked laser using a very efficient integrated-optics electro-optical waveguide modulator which has a frequency modulation bandwidth of 4 GHz. One of the sidebands generated by this modulation is locked to the cavity: this allows tuning of the locked laser by tuning the modulation frequency. This approach worked much better than the high-frequency AOM which we originally planned to use for this purpose. Substantial progress is being made in constructing a second solid-state source for the cooling radiation at 231 nm. Our original intention had been to use an injection-locked high power diode laser to "amplify" the 924 nm radiation from a low power oscillator to a level which would provide useful amounts of frequency quadrupled radiation at 231 nm. We recently discovered a source (also in Germany) of "tapered amplifiers" which work at 920 nm. This approach is a much better one than the injection-locked approach since the tapered laser works as a pure amplifier and doesn't suffer from self-oscillation which introduces practical problems with injection-locked devices. Two of these devices were purchased and they show about a 10 db gain and have proven to be easy to work with (compared to an injection-locked laser). We are building a system with two of these laser in cascade to amplify the 10 mW from the oscillator to about 1 W for subsequent frequency quadrupling (the tapered amplifiers are capable of producing 1.5 W).

Bibliography Type:	Description: (Last Updated: 03/30/2017)
Articles in Peer-reviewed Journals	Wynar RH, Iinuma M, Nagourney W, Fortson EN. "Isotope shift of the thallium (6s6p^2)^4P_{1/2} state" Phys Rev A. Sep 2003;68:32504. , Sep-2003